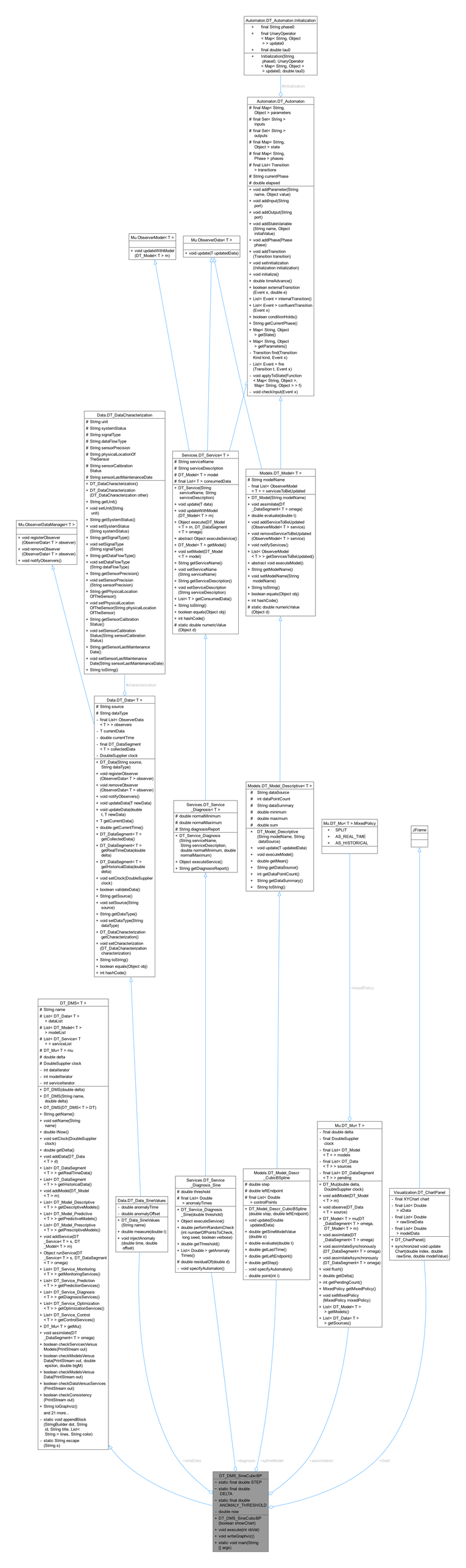
Automaton.DT_Automaton.Initialization
+ final String phase0
+ final UnaryOperator < Map< String, Object > > update0
+ final double tau0
+ Initialization(String phase0, UnaryOperator < Map< String, Object > > update0, double tau0)
#initialization
Automaton.DT_Automaton
final Map< String, Object > parameters
final Set< String > inputs
final Set< String > outputs
final Map< String, Object > state
final Map< String, Phase > phases
final List< Transition > transitions
String currentPhase
double elapsed
+ void addParameter(String name, Object value)
+ void addInput(String port)
+ void addOutput(String port)
+ void addState Variable (String name, Object initialValue)
+ void addPhase(Phase phase)
+ void addTransition (Transition transition)
+ void setInitialization (Initialization initialization)
+ void initialize()
+ double timeAdvance()
+ boolean externalTransition (Event x, double e)
+ List< Event > internalTransition()
+ List< Event > confluentTransition (Event x)
+ boolean conditionHolds()
+ String getCurrentPhase()
+ Map< String, Object > getState()
+ Map< String, Object > getParameters()
- Transition find(Transition Kind kind, Event x)
- List< Event > fire (Transition t, Event x)
- void applyToState(Function < Map< String, Object >, Map< String, Object > > f)
- void checkInput(Event x)
Mu.ObserverModel< T >
+ void updateWithModel (DT_Model< T > m)
Mu.ObserverData< T >
+ void update(T updatedData)
Data.DT_DataCharacterization
String unit
String systemStatus
String signalType
String dataFlowType
String sensorPrecision
String physicalLocationOf TheSensor
String sensorCalibration Status
String sensorLastMaintenanceDate
+ DT_DataCharacterization()
+ DT_DataCharacterization (DT_DataCharacterization other)
+ String getUnit()
+ void setUnit(String unit)
+ String getSystemStatus()
+ void setSystemStatus (String systemStatus)
+ String getSignalType()
+ void setSignalType (String signalType)
+ String getDataFlowType()
+ void setDataFlowType (String dataFlowType)
+ String getSensorPrecision()
+ void setSensorPrecision (String sensorPrecision)
+ String getPhysicalLocation OfTheSensor()
+ void setPhysicalLocation OfTheSensor(String physicalLocation OfTheSensor)
+ String getSensorCalibration Status()
+ void setSensorCalibration Status(String sensorCalibration Status)
+ String getSensorLastMaintenance Date()
+ void setSensorLastMaintenance Date(String sensorLastMaintenanceDate)
+ String toString()
#characterization
Services.DT_Service< T >
String serviceName
String serviceDescription
DT_Model< T > model
final List< T > consumedData
+ DT_Service(String serviceName, String serviceDescription)
+ void update(T data)
+ void updateWithModel (DT_Model< T > m)
+ Object execute(DT_Model < T > m, DT_DataSegment < T > omega)
+ abstract Object executeService()
+ DT_Model< T > getModel()
+ void setModel(DT_Model < T > model)
+ String getServiceName()
+ void setServiceName (String serviceName)
+ String getServiceDescription()
+ void setServiceDescription (String serviceDescription)
+ List< T > getConsumedData()
+ String toString()
+ boolean equals(Object obj)
+ int hashCode()
# static double numericValue (Object d)
Models.DT_Model< T >
String modelName
- final List< ObserverModel < T > > servicesToBeUpdated
+ DT_Model(String modelName)
+ void assimilate(DT _DataSegment< T > omega)
+ double evaluate(double t)
+ void addServiceToBeUpdated (ObserverModel< T > service)
+ void removeServiceToBeUpdated (ObserverModel< T > service)
+ void notifyServices()
+ List< ObserverModel < T > > getServicesToBeUpdated()
+ abstract void executeModel()
+ String getModelName()
+ void setModelName(String modelName)
+ String toString()
+ boolean equals(Object obj)
+ int hashCode()
# static double numericValue (Object d)
Mu.ObserverDataManager< T >
+ void registerObserver (ObserverData< T > observer)
+ void removeObserver (ObserverData< T > observer)
+ void notifyObservers()
Data.DT_Data< T >
String source
String dataType
- final List< ObserverData < T > > observers
- T currentData
- double currentTime
- final DT_DataSegment < T > collectedData
- DoubleSupplier clock
+ DT_Data(String source, String dataType)
+ void registerObserver (ObserverData< T > observer)
+ void removeObserver (ObserverData< T > observer)
+ void notifyObservers()
+ void updateData(T newData)
+ void updateData(double t, T newData)
+ T getCurrentData()
+ double getCurrentTime()
+ DT_DataSegment< T > getCollectedData()
+ DT_DataSegment< T > getRealTimeData(double delta)
+ DT_DataSegment< T > getHistoricalData(double delta)
+ void setClock(DoubleSupplier clock)
+ boolean validateData()
+ String getSource()
+ void setSource(String source)
+ String getDataType()
+ void setDataType(String dataType)
+ DT_DataCharacterization getCharacterization()
+ void setCharacterization (DT_DataCharacterization characterization)
+ String toString()
+ boolean equals(Object obj)
+ int hashCode()
Services.DT_Service _Diagnosis< T >
double normalMinimum
double normalMaximum
String diagnosisReport
+ DT_Service_Diagnosis (String serviceName, String serviceDescription, double normalMinimum, double normalMaximum)
+ Object executeService()
+ String getDiagnosisReport()
Models.DT_Model_Descriptive< T >
String dataSource
int dataPointCount
String dataSummary
double minimum
double maximum
double sum
+ DT_Model_Descriptive (String modelName, String dataSource)
+ void update(T updatedData)
+ void executeModel()
+ double getMean()
+ String getDataSource()
+ int getDataPointCount()
+ String getDataSummary()
+ String toString()
Mu.DT_Mu< T > MixedPolicy
+ SPLIT
+ AS_REAL_TIME
+ AS_HISTORICAL
JFrame
-mixedPolicy
DT_DMS< T >
String name
List< DT_Data< T > > dataList
List< DT_Model< T > > modelList
List< DT_Service< T > > serviceList
DT_Mu< T > mu
double delta
DoubleSupplier clock
- int dataIterator
- int modelIterator
- int serviceIterator
+ DT_DMS(double delta)
+ DT_DMS(String name, double delta)
+ DT_DMS(DT_DMS< T > DT)
+ String getName()
+ void setName(String name)
+ double tNow()
+ void setClock(DoubleSupplier clock)
+ double getDelta()
+ void addData(DT_Data < T > d)
+ List< DT_DataSegment < T > > getRealTimeData()
+ List< DT_DataSegment < T > > getHistoricalData()
+ void addModel(DT_Model < T > m)
+ List< DT_Model_Descriptive < T > > getDescriptiveModels()
+ List< DT_Model_Predictive < T > > getPredictiveModels()
+ List< DT_Model_Prescriptive < T > > getPrescriptiveModels()
+ void addService(DT _Service< T > s, DT _Model< T > m)
+ Object runService(DT _Service< T > s, DT_DataSegment < T > omega)
+ List< DT_Service_Monitoring < T > > getMonitoringServices()
+ List< DT_Service_Prediction < T > > getPredictionServices()
+ List< DT_Service_Diagnosis < T > > getDiagnosisServices()
+ List< DT_Service_Optimization < T > > getOptimizationServices()
+ List< DT_Service_Control < T > > getControlServices()
+ DT_Mu< T > getMu()
+ void assimilate(DT _DataSegment< T > omega)
+ boolean checkServicesVersus Models(PrintStream out)
+ boolean checkModelsVersus Data(PrintStream out, double epsilon, double bigM)
+ boolean checkModelsVersus Data(PrintStream out)
+ boolean checkDataVersusServices (PrintStream out)
+ boolean checkConsistency (PrintStream out)
+ String toGraphviz()
and 21 more...
- static void appendBlock (StringBuilder dot, String id, String title, List< String > lines, String color)
- static String escape (String s)
Data.DT_Data_SineValues
- double anomalyTime
- double anomalyOffset
+ DT_Data_SineValues (String name)
+ double measure(double t)
+ void injectAnomaly (double time, double offset)
Services.DT_Service _Diagnosis_Sine
double threshold
final List< Double > anomalyTimes
+ DT_Service_Diagnosis _Sine(double threshold)
+ Object executeService()
+ double performRandomCheck (int numberOfPointsToCheck, long seed, boolean verbose)
+ double getThreshold()
+ List< Double > getAnomaly Times()
# double residualOf(double d)
- void specifyAutomaton()
Models.DT_Model_Descr _CubicBSpline
double step
double leftEndpoint
final List< Double > controlPoints
+ DT_Model_Descr_CubicBSpline (double step, double leftEndpoint)
+ void update(Double updatedData)
+ double getSineModelValue (double x)
+ double evaluate(double t)
+ double getLastTime()
+ double getLeftEndpoint()
+ double getStep()
- void specifyAutomaton()
- double point(int i)
Mu.DT_Mu< T >
- final double delta
- final DoubleSupplier clock
- final List< DT_Model < T > > models
- final List< DT_Data < T > > sources
- final List< DT_DataSegment < T > > pending
+ DT_Mu(double delta, DoubleSupplier clock)
+ void addModel(DT_Model < T > m)
+ void observe(DT_Data < T > source)
+ DT_Model< T > mu(DT _DataSegment< T > omega, DT_Model< T > m)
+ void assimilate(DT _DataSegment< T > omega)
+ void assimilateSynchronously (DT_DataSegment< T > omega)
+ void assimilateAsynchronously (DT_DataSegment< T > omega)
+ void flush()
+ double getDelta()
+ int getPendingCount()
+ MixedPolicy getMixedPolicy()
+ void setMixedPolicy (MixedPolicy mixedPolicy)
+ List< DT_Model< T > > getModels()
+ List< DT_Data< T > > getSources()
Visualization.DT_ChartPanel
- final XYChart chart
- final List< Double > xData
- final List< Double > rawSineData
- final List< Double > modelData
+ DT_ChartPanel()
+ synchronized void update Chart(double index, double rawSine, double modelValue)
~sineData
~diagnosis
~splineModel
~assimilation
~chart
DT_DMS_SineCubicBP
~ static final double STEP
~ static final double DELTA
~ static final double ANOMALY_THRESHOLD
- double now
+ DT_DMS_SineCubicBP (boolean showChart)
+ void execute(int nbVal)
+ void writeGraphviz()
+ static void main(String [] args)